# Project “Plantimeter”

## Description

## 1 Topic

The company PLANTS IN FOCUS S.A. presents its latest product – *Plantimeter*. It is a comprehensive platform for the automated care of house and garden plants. Every plant has different requirements. The *Plantimeter* system ensures that your plants lack nothing. Specifically, the parameters of the soil, the air and the lighting conditions are measured. If necessary, the system irrigates the plant and provides light. *Plantimeter* is ideal for use in the greenhouse or indoors. The system can be installed by the buyer himself. The system requires a Wifi network to enable communication between the components of the system.

Since you are the local computer scientist, the company will turn to you to implement the project Plantimeter according to the here presented details.

## 2 General project description

The system includes a base station, also known as a *hub*, which is mounted at a central location at the site. Preferably on a wall. The hub is supported by plant stations called nodes, which are mounted directly on the respective plants.

### 2.1 Hub

The base station offers an interactive touch interface, which clearly displays the most important data of the plants in our system. Here, the current measured values of the plants can be recorded and the nodes can be controlled. The environment is specially designed for small displays and can also be accessed on a smartphone.

The hub stores and analyzes the measurement data from the nodes and decides whether a plant needs to be watered or the grow light should be turned on. The user can also control the plant stations directly. For example, you can request the watering of a plant.

*Illustration 1 - Hub Example*

## 2.2 Node

The nodes are equipped with a wide variety of sensors and controls. These enable us to determine data about the environment of the plants. For example, the moisture of the soil is measured to determine whether the plant has enough water available. The nodes send this data to their hub at regular intervals. The nodes perform desired actions on command, such as watering the plant.

*Illustration 2 – Node, for example, without connections for water and light.*

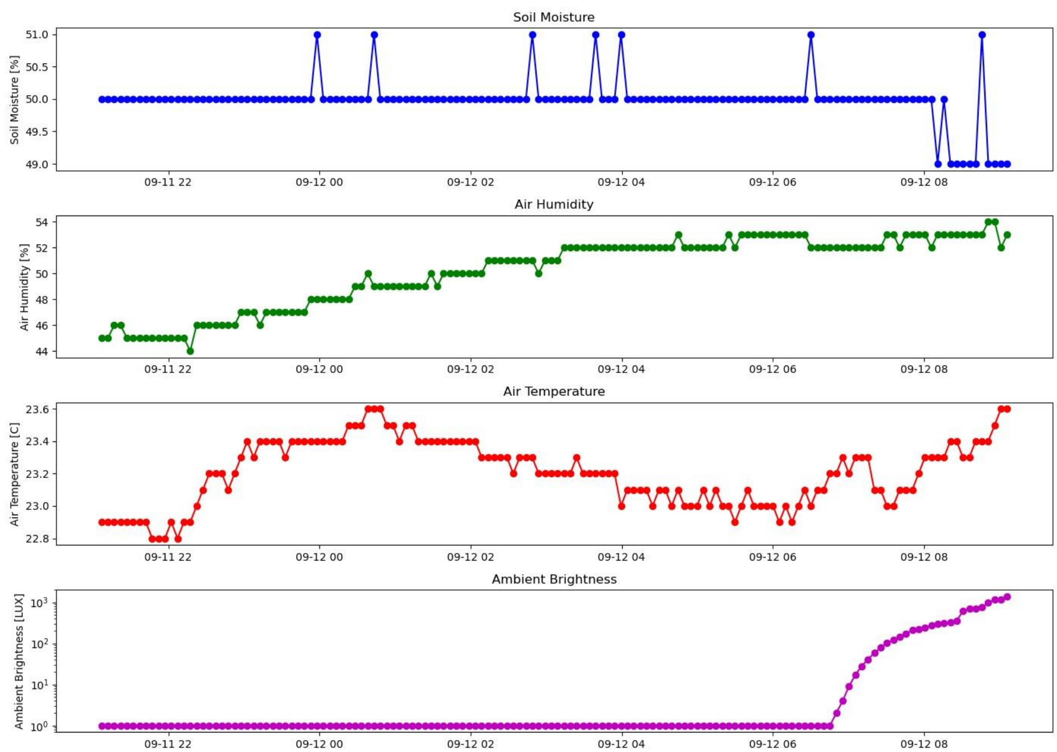


*Illustration 3 – Example, measurement data of a node*

## 2.3 Website

The website is the counterpart to the hub's touch interface. Here the user can configure the system. Here, too, the measurement data of the plants can be viewed, filtered and analysed in a more targeted manner. Another feature is the ability to view the logs of the nodes for errors and events.

### 2.4 Firmen Server

The company server is available to back up customer data in case the hub's storage fails. Customer data is uploaded here at regular intervals.

| Week | from | until | Labor | Workshop | Presentations |
|---|---|---|---|---|---|
| *45* | 04/11/24 | 10/11/24 | L1 Raspberry Pi | A1 Getting Started | |
| *46* | 11/11/24 | 17/11/24 | | | |
| *47* | 18/11/24 | 24/11/24 | L2 Database | A2 Irrigation | |
| *48* | 25/11/24 | 01/12/24 | | | |
| *49* | 02/12/24 | 08/12/24 | L3 Touch Interface<br>To plan | A3 Bodenmessung | |
| *50* | 09/12/24 | 15/12/24 | | | |
| *51* | 16/12/24 | 22/12/24 | L4 Website | A4 Luftmessung | |
| | Christmas holidays | | | | |
| *02* | 06/01/25 | 12/01/25 | L4 Website | A4 Luftmessung | |
| *03* | 13/01/25 | 19/01/25 | | A5 Light Metering | **REPIF1*** |
| *04* | 20/01/25 | 26/01/25 | | | |
| *05* | 27/01/25 | 02/02/25 | | A6 Node | |
| *06* | 03/02/25 | 09/02/25 | | | |
| | Carnival holidays | | | | |
| *09* | 24/02/25 | 02/03/25 | L5 Network | A6 Node | |
| *10* | 03/03/25 | 09/03/25 | L6 Server | | |
| *11* | 10/03/25 | 16/03/25 | | A7 Lighting<br>Planning | |
| *12* | 17/03/25 | 23/03/25 | | | |
| *13* | 24/03/25 | 30/03/25 | L7 Node<br>To plan | | |
| *14* | 31/03/25 | 06/04/25 | | | |
| | Easter holidays | | | | |
| *17* | 21/04/25 | 27/04/25 | LX Improvements | AX Improvements | **REPIF2*** |
| *28* | 28/04/25 | 04/05/25 | | | |
| *19* | 05/05/25 | 11/05/25 | | | |
| *20* | 12/05/25 | 18/05/25 | | | |
| *21* | 19/05/25 | 25/05/25 | | | |
| | Pentecost holidays | | | | |
| *23* | 02/06/25 | | | | **Project*** |

** This information is for temporal guidance only and is not definitively determined. The teacher will set the exact dates and communicate them to the class.*

## 3 Database

In addition to the website, the database is the heart of the project. This is where everything comes together. The database can be set up locally for test purposes, for example with Xampp or Uniserver. Your SQL script must be able to run flawlessly on a MySQL-based database server.

The database stores data on the following units:

| Units | Explanation |
|---|---|
| **User** | A user (eng. Operator) is a person who can log in to the system. The login to the website is done by **e-mail** in combination with a **password**.<br>A user also has a **first name** & **last name**. |
| **Nodes** | A node is responsible for a maximum of one **plant species**. Each node can have a **name**. If a node is planted, it contains the **date** on which it was assigned to the plant species. Each node has an **owner**. The default owner of a node is the user "admin@ecample.com" (id=1). For unique identification, each node has a **MAC address**. |
| **Plant species** | A plant species (eng. Variety) is defined by its **common name**. A **botanical name** can also be given. To optimally supply plants with light and water, we store the optimal threshold values for **brightness** and **soil moisture for each plant species**.<br>Guideline values for **air temperature** and **humidity** can also be stored. |
| **Measurements** | A measurement contains various sensor data that captures the following values:<br><br>• **Soil moisture** (indicates the moisture in the soil)<br>• **Air temperature** (detects the temperature of the environment)<br>• **Humidity** (measures the water vapor content in the air)<br>• **Brightness (lumens)** (indicates light intensity)<br>• **State of the light** (shows in percentage how much the node's grow light is set)<br>In addition, the **timestamp is** recorded, which indicates the exact time of the measurement, as well as the **node affiliation**, which indicates which node the data comes from. If a measurement cannot be read or is outside a reasonable range, it is stored as NULL. |
| **Tasks** | The system & users give up tasks that the **nodes** must do. A task corresponds to either **lighting** or **watering**. Values **between** 0-500 can be specified as quantities. Each task has a **due date and time**, which is "now" by default. New tasks are always "**not done**" by default. A task is marked as complete once it is completed. |